Supplementary data 2: Table 1. Characteristics of studies investigating rehabilitation interventions for survivors of cardiac arrest

Study	Title	Population	Rehabilitation intervention	Outcomes		Summary of results
Author, year, country **Study design** **Setting**		**a. Participants** **b. Number of included participants (N)** **c. Age (mean)** **d. Gender (male, %)** **e. Ethnicity (% Caucasian)**	**a. Description** **b. Duration (time period and/or number of sessions)**	**Outcomes of interest**	**Follow-up period**	
Burke, 2005, USA Retrospective chart review Freestanding rehabilitation hospital	Rehabilitation outcomes of cardiac and non-cardiac anoxic brain injury: a single institution experience	a. Acquired brain injury due to cardiac arrest. b. n=13 c. 52.5 d. 54 e. 61	a. Comprehensive, multi-disciplinary inpatient rehabilitation services. b. Admission period mean 69.8 days (SD 59.4)	FIM subscales activities of daily living, mobility, cognition and total	Admission, discharge	Total FIM improved from admission mean 63.92 days (SD 32.82) to mean discharge 90.46 days (SD 41.37).
Cowan, 2001, USA Randomized controlled trial Outpatient clinic	Psychosocial nursing therapy following sudden cardiac arrest: impact on two-year survival.	a. Out-of-hospital cardiac arrest survivors b. n=66 (intervention) n=67 (control) c. NS d. 73 e. 90	a. Three components: 1) physiologic relaxation, 2) Cognitive behavioral therapy and 3) Health education focusing on cardiac risk factors. Delivered by experienced nurses. Control received only the health education component. b. 4 weeks, mean 11 sessions (30 minutes each)	All-cause mortality; Risk of cardiovascular death; Non-Fatal cardiac effects	2 years	Reduction in risk of all-cause mortality for the intervention group was 62%, but this was not statistically significant, (p=0.13). Risk of cardiovascular death was significantly reduced in the intervention group by 86% compared to conventional treatment at two-years follow-up (hazards ratio =0.14; p=0.03; one death in the intervention group due to stroke, six out of seven deaths in control group due cardiac arrest. Confidence intervals for these results were not available).
Dougherty, 2004, 2005, USA Randomized controlled trial Telephone	Short-term efficacy of a telephone intervention by expert nurses after an ICD; Long-Term Outcomes of a Telephone Intervention After an ICD	a. Cardiac arrest survivors with first time ICD implantation for secondary prevention of cardiac arrest b. n=38 (intervention), n=27 (control) c. 63.5 d. 74 e. 91	a. 1) Booklet mailed to study participants on strategies to manage recovery, 2) structured information provided by experienced cardiovascular nurses to improve self-efficacy to deal with illness demands, and to reduce anxiety, through identification of illness related problems and behavioral strategies to manage them including role playing, problem-solving, goal-setting and collaborating on the learning assignment for the coming week. Usual care participants received treatment as usual from their health care providers and standardized hospital-based education about the ICD in the form of a booklet, videotape, or both.	SF-12 Physical + Mental component sub-scale (separate results for other outcomes reported in the paper were not available for CA-survivors)	Baseline, 1, 3, 6, 12 months	Rehabilitation interventions showed no statistical effectiveness on either SF-12 physical or mental subscales at any follow-up point compared to standard care.

			b. First 8 weeks after hospital discharge and ICD implantation (15-20 minute calls, number of calls NS).			
Dougherty, 2008, USA A single group pre-post test design Outpatient group exercise	Aerobic exercise improves fitness and heart rate variability after an ICD	a. Cardiac arrest survivors with first time ICD implantation for secondary prevention of cardiac arrest b. n=10 c. 54.8 d. 90 e. NS	a. Supervised aerobic exercise plus home walking b. 3 days per week for total of 8 weeks (24 sessions) + 1 hour of home walking twice a week.	SF-12 Physical + Mental component sub-scale; State Trait Anxiety Inventory; Center for Epidemiological Studies–Depression Scale; Total exercise time (minutes); Oxygen pulse (VO2/HR); Metabolic equivalent of task; RT3 accelerometer	Baseline, 8 weeks, 6 months	Quality of life (Short Form–12) showed a non-significant improvement in physical and mental sub-scale scores and non-significant reduction in anxiety and depression. Exercise duration, oxygen uptake at anaerobic threshold, and metabolic equivalents were improved after 8 weeks of exercise. There were no lethal cardiac arrhythmias experienced during exercise testing and no participants required cardioversion. There were no sustained ventricular arrhythmias during supervised exercise or home walking sessions in any of the study subjects. SF-12 physical health were sustained at 6 months as well as an increase in daily activity as measured by RT3 accelerometer.
Dougherty, 2019, USA Randomized controlled trial Telephone	Patient plus partner trial: A randomized controlled trial of 2 interventions to improve outcomes after an initial ICD	a. Cardiac arrest survivors with first time ICD implantation for secondary prevention of cardiac arrest b. n=66 c. 62.3 d. 72 e. 94	a. Intervention consisted of 4 elements: 1) Information booklet with strategies for health recovery after ICD implant. 2) Nurse telephone support to improve self-efficacy and problem solve. Components of this support were as per the intervention in Dougherty, 2004, 2005. 3) Pager access to a study nurse 4) an informational video provided by the device company. b. 10 telephone calls over 12 weeks	SF-12 Physical + Mental component sub-scale; State Trait Anxiety Inventory; Patient health questionnaire-9	Baseline, 3, 6, 12 months	All outcomes improved between baseline and 12 months follow-up. No effect sizes available.
Fertl, 2000, Austria Retrospective chart review Inpatient neurological rehabilitation	Neurological rehabilitation of severely disabled cardiac arrest survivors. Part I. Course of post-acute inpatient treatment	a. Out-of-hospital cardiac arrest survivors who suffered anoxic brain injury and required prolonged intensive care treatment b. n=20 c. 47.6 d. 85 e. NS	a. Daily multidisciplinary neurological rehabilitation. Speech therapists and psychologist were available for those needing their service. Admission period mean 84 days (SD 57) (minimum 15 sessions per week of b. physical and occupational therapy)	Barthel Index	Admission + discharge	Mean improvement in Barthel index 3.4 (SD 4.4)

Howell, 2013, Germany Retrospective cohort study Inpatient neuro rehabilitation center	Rehabilitation outcome of anoxic-ischaemic encephalopathy survivors with prolonged disorders of consciousness	a. Cardiac arrest survivors, direct transfer from intensive care unit, all in coma, vegetative state or minimally conscious state b. n=113 c. 55 d. 74 e. NS	a. Daily neurorehabilitation b. Mean 84 days (SD 50)	Glasgow outcome scale; FIM; Coma remission scale	Admission + discharge	Total FIM improved from mean (SD) 18 (0) to 25 (19) points between admission and discharge. 6.2% of patient achieved a good functional outcome defined as Glasgow Outcome Scale 4-5. Coma remission scale improved from mean (SD) change of 9 (5) to 13 (7) points (scale 0-24, higher score indicates better recovery).
Kim, 2014, South Korea Retrospective review of medical records Outpatient hospital-based	Cardiac rehabilitation after acute MI resuscitated from cardiac arrest	a. Cardiac arrest survivors who received successful percutaneous coronary intervention for acute MI b. n= 8 c. 46.8 d. 88 e. NS	a. Cardiac rehabilitation including aerobic exercise, advice on secondary risk factors, diet and lifestyle, advice on medication by a cardiologist. Exercise was continued at home at 60% intensity of the heart rate reserve. b. 6 weeks (3x50-minute per week exercise programs for 6 weeks)	Cardiovascular-related complications during exercise monitoring; Peak oxygen consumption (VO2peak (mL/kg/min)); Exercise duration (minutes)	Baseline + 8 weeks	Significant improvement in exercise capacity. No fatal cardiac complications, such as abnormal ECG, cardiac arrest, death or myocardial infarction observed.
Kim, 2016, USA Prospective, pre-post single group experimental design Telephone	An intervention for cardiac arrest survivors with chronic fatigue: A feasibility study with preliminary outcomes	a. Cardiac arrest survivors at least 3 months post cardiac arrest with chronic fatigue b. n=8 c. 53.2 d. 56 e. 100	a. Energy Conservation and Problem Solving Therapy delivered by an occupational therapist. b. Up to 4 weeks (45 minute sessions twice a week)	Modified Fatigue Impact Scale; Fatigue Severity Scale; Patient Reported Outcomes Measurement Information System-Fatigue scale	Pre-test, post-test (range 3-5 weeks)	Significant decreases on the Modified Fatigue Impact Scale total ($p<0.001$), subscales physical ($p=0.001$) and cognitive ($p = 0.006$) fatigue were observed with small to moderate effect sizes of $r=0.23–0.25$. Change effect sizes were small for the Fatigue Severity Scale ($r=0.11$), Patient Reported Outcomes Measurement Information System-Fatigue scale ($r=0.19$).
Mion, 2019, UK Prospective cohort study In-patient and outpatient clinic	Care After Resuscitation: Implementation of the United Kingdom's First Dedicated Multidisciplinary Follow-Up Program for Survivors of Out-of-Hospital Cardiac Arrest	a. Cardiac arrest survivors with good neurological recovery, Cerebral Performance Scale Category 1-2 n=19 b. 61 c. 84 d. NS e.	a. Inpatient information provided via leaflets, bespoke video and direction to a social media website for cardiac arrest survivors and caregivers; telephone and clinic follow-up with ICU nurse, cardiologist and psychiatrist. If psychological issues were identified, patients and caregivers were offered further interventions. b. In-hospital + clinic follow-up at: 8-weeks, 6-months and 12-months post-hospital discharge.	SF-36 physical and mental domain scores	Baseline, 6 months	Significant improvement in all domains of Short-form 36 (except general health) at 6 months.
Moulaert, 2015, The Netherlands	Early neurologically-focused follow-up after cardiac arrest improves quality of	a. Survivors of cardiac arrest at least two weeks after event, living at home b. n=97 (Intervention) n=98 (Control)	a. Intervention for survivors of cardiac arrest and their caregivers provided by specialist nurses including 1. Screening for cognitive and emotional problems. 2. Provision of support and information on cardiac arrest	SF-36 domain scores; EuroQol Visual Analogue Scale; Cognitive log	Baseline, 3 + 12 months	At 12 months there were significant differences in estimated means in favour of the intervention group on three domains of quality of life on the SF-36: Role Emotional

Multicenter randomised controlled trial At clinic or at home	life at one year: A randomised controlled trial	c. 60 (Intervention) 69 (Control) d. 83 (Intervention) 84 (Control) e. NS	and possible neurological consequences. 3. Promotion of self-management strategies. 4. Referral to specialized care if indicated. Control group received standard care with potential for referral to cardiac rehabilitation. b. 1-6 individual sessions	Adult Memory and Information processing battery task A; Verbal fluency; Trail making Test A; Trail making Test B; Paragraph recall direct; Paragraph recall delayed; Cognitive Failures Questionnaire; Hospital Anxiety and Depression Scale (anxiety and depression sub-scales and total); Impact of Event Scale		(p=0.006), Mental Health (p=0.003) and General Health (p=0.010). No significant effectiveness on cognitive tests compared to standard care at any follow-up point. The intervention group scored significantly better on overall emotional state (anxiety and depression) and anxiety at one year.
Schmidt, 1997, USA Retrospective chart review In-patient rehabilitation unit	Anoxic encephalopathy: Outcome after inpatient rehabilitation	a. Patients admitted to a rehabilitation unit with cerebral anoxia (15 due to cardiac arrest and 11 for other causes) b. n=26 c. 58 d. 66 e. NS	a. In-patient rehabilitation unit b. Admission period mean 59.5 days (SD 41.4)	FIM	Admission + discharge	Total FIM improved from admission mean 54.15 (SD 29.85) to discharge mean 81.23 (SD 26.34).
Shah, 2007, USA Retrospective chart review Freestanding rehabilitation hospital	A comparison of functional outcomes in hypoxia and traumatic brain injury: A pilot study	a. Survivors of cardiac arrest who suffered anoxic brain injury b. n=15 c. 50.8 d. 60 e. 87	a. In-patient rehabilitation b. Mean 61.2 days (SD 68.4)	FIM subscales activities of daily living, mobility, cognition and total	Admission + discharge	Total FIM improved from admission mean 61.93 (SD 33.23) to discharge mean 94.67 (SD 37.34)
Tazopoulou, 2016, France Observational single cohort study Residential care	Rehabilitation following cerebral anoxia: An assessment of 27 patients	a. Adults with cerebral anoxia in residential care (11 due to cardiac arrest, 9 due to other cause) b. n=20 c. 46 d. 70 e. NS	a. Psychotherapy, support group, physical activities and cultural and/or artistic activities. Participants could choose to be in all or some of the activities. b. 2-months	Glasgow outcome score extended; Bermont Vost Alexithymia questionnaire; Patient Competency Rating scale (agnosia); Quality of Life After Brain Injury questionnaire; Barrow Neurological Institute screen of higher cerebral functions; Hospital Anxiety and Depression scale	Baseline, 2 + 4 months	Quality-of-life was significantly improved between baseline and intervention end at two months. No change found in neurological function or anxiety and depression.

ICD: Implantable Cardioverter-Defibrillator; FIM: Functional Independence Measure; NS: Not stated; SF: Short form health survey